III. Central projections

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Motivation. Given $P, Q \in \mathcal{P}(A)$, must we have either $P \preceq Q$ or $Q \preceq P$?
We will see that in general this isn't the case, but that it is when
$$\mathrm{Cen}(A) = \mathbb{C} \cdot I_H$$
For example, this holds in $B(H)$, as also noted in above example.

We define the <u>centre</u> of a von Neumann alg. A as
$$\mathrm{Cen}(A) = A \cap A' = \{ T \in A \,;\, TS = ST \ \forall S \in A \}$$
which is a commutative C^*-subalgebra of A that is weakly closed in A (since A' is), so it's a commutative von Neumann algebra.

<u>Central projections</u> are $\mathcal{P}_c(A) = \mathcal{P}(A) \cap \mathrm{Cen}(A)$.

Proposition. Let $P, Q \in \mathcal{P}(A)$.

(i) $P \sim_A Q \Rightarrow PZ \sim_A QZ, \ \forall Z \in \mathcal{P}_c(A)$

↳ $UPU^* = Q \Rightarrow UPZU^* = UPU^*Z = QZ$

(ii) $P \preceq Q \Rightarrow PZ \preceq QZ, \ \forall Z \in \mathcal{P}_c(A)$.

↳ $R \sim_A P \overset{(i)}{\Rightarrow} RZ \sim_A PZ$

For $RZ \leq QZ$, note $\mathrm{im}(RZ) \subset \mathrm{im}(QZ)$.

<u>Def.</u> If $P \in \mathcal{P}(A)$, the <u>central support</u> of P is $Z(P) := \bigwedge \{ Q \in \mathcal{P}_c(A) \,;\, P \leq Q \}$

i.e. the infimum of all central projections which are greater than P.

($Z(P)$ is the smallest central proj., dominating P.)

<u>Prop.</u> If $P \in \mathcal{P}(A)$, then

$Z(P)$ projects onto $\overline{\sum_{T \in A} \mathrm{im}(TP)}$.

<u>Pf.</u> Let $Q = \bigvee_{T \in A} P_{\mathrm{im}(TP)}$. WTS: $Z(P) = Q$.

- Since $\mathrm{im}(Q) = \overline{\sum_{T \in A} \mathrm{im}(TP)}$ is invariant for all operators $T \in A$ (and also $T^* \in A$), we have

  $QT = TQ$, thus $Q \in \mathrm{Cen}(A)$.

  Trivially, $\mathrm{im}(P) \subset \mathrm{im}(Q)$ so $P \leq Q$, implying $\boxed{Z(P) \leq Q.}$

- For the other inequality, note $Z(P) \in \mathrm{Cen}(A)$

  i.e. $Z(P)T = TZ(P) \quad \forall T \in A$

  so $\mathrm{im}(Z(P))$ is invariant $\forall T \in A$.

  But $P \leq Z(P)$ implies $\mathrm{im}(P) \subset \mathrm{im}(Z(P))$ and so $T(\mathrm{im}(P)) \subset \mathrm{im}(Z(P))$, thus

  $P_{\mathrm{im}(TP)} \leq Z(P) \quad \forall T \in A$

and by def. of Q, it follows that $\boxed{Q \leq z(P)}$.

$\blacksquare$

<u>Lemma</u>. Let $P, Q \in \mathcal{P}(A)$. TFAE:

(i) $z(P)z(Q) \neq 0$ (P and Q aren't "centrally orthogonal")

(ii) $\exists\, P_1, Q_1 \in \mathcal{P}(A) \setminus \{0\}$ s.t.

$$P_1 \leq P, \quad Q_1 \leq Q, \quad \text{and} \quad P_1 \sim_A Q_1$$

<u>Pf.</u> $(i \to ii)$

By previous prop., $\underbrace{\left(\bigvee_{T \in A} P_{\mathrm{im}(TP)}\right)}_{z(P)} \cdot \underbrace{\left(\bigvee_{S \in A} P_{\mathrm{im}(SQ)}\right)}_{z(Q)}$ $\neq 0$, so now $\exists\, v, w \in H$ and $T, S \in A$ s.t.

$$\langle TPv, SQw \rangle = \langle Q\underbrace{S^*T}_{\in A}Pv, w \rangle \neq 0.$$

So now $Q(S^*T)P \neq 0$ and we define

$$P_1 := P_{P(S^*T)^*Q}, \quad Q_1 := P_{Q(S^*T)P}$$

$\Rightarrow P_1, Q_1 \neq 0$, $P_1 \leq P$, $Q_1 \leq Q$ and $P_1 \sim Q_1$.

look at image

Corollary from before

$(ii \to i)$

Suppose $V^*V = Q_1$, $VV^* = P_1$.

$\Rightarrow PVQ \neq 0$ since $\exists\, v \in \mathrm{im}(Q_1) \setminus \{0\}$. } Apply PVQ to v...

$\Rightarrow PVQ \cdot z(P)z(Q) = (\underbrace{z(P) \cdot P}_{=P}) V (\underbrace{z(Q) \cdot Q}_{=Q}) = PVQ \neq 0$

$z(P), z(Q) \in \mathrm{Cen}$

So trivially, $z(P)z(Q) \neq 0$. $\blacksquare$

Theorem. (The comparison theorem)

If $P, Q \in \mathcal{P}(A)$, then $\exists\, z \in \mathcal{P}_c(A)$ s.t.

$$Pz \preceq Qz \quad \text{and} \quad Q(I-z) \preceq P(I-z).$$

Corollary. If $\mathrm{Cen}(A) = \mathbb{C} I_H$ (we say that A is a "factor"), then $\forall P, Q \in \mathcal{P}(A)$:

$$P \preceq Q \quad \text{or} \quad Q \preceq P,$$

i.e. $\preceq$ is a total order on $\mathcal{P}(A)/\sim$.

Pf. (of Corollary)

By comparison thm., $\exists\, z \in \mathcal{P}_c(A)$ as above.
But since $\mathrm{Cen}(A) = \mathbb{C} I_H$, we get
$z = I$ (i.e. $P \preceq Q$) or $z = 0$ ($Q \preceq P$).

Pf. (of theorem)

Let $(P_\alpha)_{\alpha \in I}$ and $(Q_\alpha)_{\alpha \in I}$ be families of operators s.t. $P_\alpha \leq P$, $Q_\alpha \leq Q$, $P_\alpha \sim Q_\alpha$.
The set $\{(P_\alpha, Q_\alpha) ; \alpha \in I\}$ is non-empty and every chain admits an upper bound (i.e. the sum $(\bigvee_\beta P_\beta, \bigvee_\beta Q_\beta)$), so applying Zorn's lemma gives a maximal element (P_0, Q_0) s.t. $P_0 \sim Q_0$ and $P_0 \leq P$, $Q_0 \leq Q$.
If $P_0 = P$ or $Q_0 = Q$, we are done.

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If not, consider $P - P_0$ and $Q - Q_0$.

↳ These do not have subprojections

$P_\alpha \leq P - P_0$, $Q_\alpha \leq Q - Q_0$ with $P_\alpha \sim Q_\alpha$

otherwise the pair (P_0, Q_0) isn't maximal since then $P - P_\alpha \geq P_0$, $Q - Q_\alpha \geq Q_0$.

By the lemma above, $z(P-P_0)\cdot z(Q-Q_0) = 0$.

Take $z = z(Q - Q_0)$. Then

$$z(P-P_0)\cdot z = 0 \underset{\substack{\text{dominates} \\ P-P_0}}{\overset{z(P-P_0)}{\Longrightarrow}} (P-P_0)\cdot z = 0$$

Hence $Pz = P_0 z \underset{\substack{\uparrow \\ \text{proposition from beginning}}}{\sim} Q_0 z \leq Qz$.

Likewise, $(Q - Q_0)\cdot z \underset{\substack{\uparrow \\ Q-Q_0 \leq z}}{=} Q - Q_0$

$$\Rightarrow Q(1-z) = Q_0(1-z) \underset{\substack{\uparrow \\ \text{prop. from beginning}}}{\sim} P_0(1-z) \leq P(1-z). \quad \blacksquare$$